Education and Income

Micro-Project 1

github.com/mascon0423/ANA500

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Problem Statement

- This project investigates how income category varies with educational attainment in the Adult census sample.
- Question: Is the share earning above $50,000 higher among adults with a bachelor's degree or higher than among adults with less education?
- The outcome is an annual income category: at or below $50,000 versus above $50,000. Exact salaries are unavailable.
- Micro-Project 1 establishes the question and prepares the data for a later comparison.

Hypothesis Formulation

- Hypothesis: The bachelor's-or-higher group has a greater share of adults earning above $50,000 than the group below a bachelor's degree.
- Possible explanation: Higher educational attainment may provide access to occupations with higher earnings. This is a proposed explanation, not a finding.
- Bachelor's or higher includes Bachelors, Masters, Prof-school, and Doctorate. All other education categories form the comparison group.
- The hypothesis remains untested. Age, occupation, and work hours could also influence the observed relationship.

Acquire

- Source: UCI Adult dataset, created by Barry Becker and Ronny Kohavi from the 1994 Census database.
- Supplied file: adult income.csv. It contains 32,561 records and 15 columns: 14 attributes plus the income label.
- The supplied records match UCI's training split. The separate test split is not used in this micro-project.
- Pandas loads the local CSV as a DataFrame. One row represents an adult record. The original file is preserved.
- Key fields are education, education-num, and income. Age, work hours, and occupation provide context. fnlwgt is a survey weight, not a person ID.
- Reference: Becker & Kohavi (1996), Adult. https://doi.org/10.24432/C5XW20

Prepare: Initial Exploration

- Income categories: 24,720 records at or below $50,000 and 7,841 above $50,000. These are sample counts.
- Missing workclass: 1,836. Missing occupation: 1,843. Missing native-country: 583.
- 2,399 records contain at least one missing value. The original marker is ?. Education and income have no missing values.
- 24 profiles repeat beyond their first occurrence. Without a person identifier, identical values do not prove duplicated people.
- NumPy summaries: age ranges from 17 to 90, with median 37. Work hours range from 1 to 99 per week, with median 40.

Prepare: Cleaning Decisions

- Trim text and replace ? or empty cells with explicit missing values. Retain unknown categories without guessing replacements.
- Convert six numeric fields to integers and nine text fields to categorical data. Validate income labels and education-code agreement.
- Retain all records, including repeated profiles. Document duplicates for later sensitivity checks rather than assuming they are errors.
- Preserve valid zero capital gains and losses. Retain plausible extreme ages and work hours rather than automatically clipping them.
- Create education_group with NumPy and encode income_gt_50k as 1 for above $50,000 and 0 otherwise. Preserve every original field.

Prepare: Readiness and Scope

- Prepared dataset: 32,561 records and 17 columns. No rows removed. Two derived fields added.
- Education groups: 8,067 bachelor's or higher, and 24,494 below bachelor's. Income rates by group have not been compared.
- Checks passed for the schema, required values, integer fields, plausible ranges, income categories, and education-code consistency.
- Unknown ancillary categories remain documented. The data are ready for the planned education comparison, subject to later analytical decisions.
- Limitations: historical 1994 data, broad income categories, unweighted sample summaries, and no causal design.
- Micro-Project 1 ends at Prepare. The hypothesis remains untested; Analyze, Report, and Act are reserved for later work.